

Exam-like problems

Q1 Hex ^{0x3FA} to decimal

```
.global main
main:
# 3, 15, 10 = 3FA
li t0, 3
li t1, 15
li t2, 10
li t3, 256 # multiply with 3
mul t0, t0, t3
li t3, 16 # multiply with F(15)
mul t1, t1, t3
add t0, t0, t1
add t0, t0, t2
mv a0, t0
ret
```

Q3 Decimal (231) to Binary

```
.global main
main:
li t0, 231
li t1, 0
li t2, 1
li t3, 0
dec to bin:
add t1, t1, t3
addi t2, t2, 1
done:
mv a0, t0
ret.
```

Q2 Binary (0b10110) to decimal

```
.global main
main:
li t0, 0b10110
li t1, 0
li t2, 1
li t3, 0
bintodec:
mul t4, t3, t2
add t1, t1, t4
mul t2, t2, 2
done:
ret
```

Q4

The problem is within the for loop, the equation $result = result * 10 + s[i];$ does not account for ASCII, so have to convert with ASCII value.
 $result * 10 + (s[i] - '0')$